Rex Chao

925-357-7186 | rexchao1@gmail.com | linkedin.com/in/rexchao1 | github.com/rexchao1
Los Angeles, CA 90012

Education

University of California-Los Angeles Los Angeles, CA
Bachelor of Science (B.S.) in Computer Science *Expected Jun. 2029*

- **Relevant Coursework:** Data Structures and Algorithms (CS 32), Computer Organization (CS 33), Object-Oriented Programming in C++ (CS 31), Linear Algebra (Math 33A), Multivariable Calculus (Math 32A/32B), Differential Equations (Math 33B)

Experience

Co-Founder & Software Engineer May. 2026 – Present
Tooey *Danville, CA*

- Co-founded and built a restaurant operations platform that forecasts per-item demand and turns those forecasts into daily prep and inventory recommendations.
- Benchmarked three demand forecasting models over a held-out year of public sales data; the gradient-boosted model won, beating a same-weekday-average baseline by 10%.
- Engineered the full stack: FastAPI backend, PostgreSQL data layer, cloud infrastructure, and simple UI.

Software Engineer Intern Jun. 2026 – Sep. 2026
Photon Commerce *San Francisco, CA*

- Debugged a custom MCP connector ahead of its production launch, reproducing failures and tracing root causes through authentication flows and unhandled edge cases.
- Simplified onboarding by cutting steps out of the account sign-up flow, reducing the work required to connect a new user.
- Researched competing fintech and document-processing platforms to inform the product roadmap.

Projects

Automated Dental Front Desk Assistant May. 2026 – Present

- Built and deployed a local AI agent now in daily use at a dental office, working 10 insurance payer portals a day and drafting 20 patient letters a week.
- Ran inference locally through Ollama models to keep patient data 100% on-premise.
- Speeds up monotonous tasks to do with calling insurance representatives and traversing their apps, while also assisting doctors with letter writing.

Fetch | *Tauri, Rust, React, TypeScript* Sep. 2026

- Built and shipped a macOS desktop app that streams web video straight from any streaming URL.
- Engineered an HLS proxy that finds `.m3u8` playlists in a page's network traffic, copying credentials by rewriting every segment URL behind one local address, and refetches signed playlists before their tokens expire.

Laoshu | *Swift, SwiftUI, SQLite* Sep. 2026 – Present

- Built an iPhone app for learning Chinese vocabulary with a spaced-repetition scheduler, an adaptive placement test, and streak tracking, using HSK word lists compiled into SQLite.
- Made the vocabulary catalogue, scheduler, and session logic into a pure Swift package, running totally local on the iOS device.

Software Factory | *Go, TypeScript, Docker* Aug. 2026 – Present

- Total developer workflow that optimizes the blend between agent-developer planning and offline autonomous agentic work using my own engineered light and dark factories.
- Involves automatic task queue and a planning layer with roadmaps, specs, and a critic pass, plus a web console for tracking all work.

Technical Skills

Languages: Python, C++, SQL
Developer Tools: FastAPI, PostgreSQL, SQLite, Docker, Git, GitHub Actions